

Vernier Calipers & Screw Gauge — 100 Problems

Nine Education IIT Academy | Class 11 Physics | Vernier: LC = 0.01 cm · Screw Gauge: pitch = 1 mm, 100 divisions, LC = 0.01 mm (unless stated otherwise)

Vernier — Readings

1. MSR = 1.2 cm, 4th division coincides.
2. MSR = 2.5 cm, 7th division coincides.
3. MSR = 0.8 cm, 9th division coincides.
4. MSR = 3.4 cm, 2nd division coincides.
5. MSR = 5.6 cm, 8th division coincides.
6. MSR = 0.5 cm, 3rd division coincides.
7. MSR = 6.2 cm, 6th division coincides.
8. MSR = 4.1 cm, 1st division coincides.
9. MSR = 2.0 cm, 5th division coincides.
10. MSR = 7.3 cm, 9th division coincides.
11. MSR = 1.9 cm, 6th division coincides.
12. MSR = 3.0 cm, 2nd division coincides.
13. MSR = 8.4 cm, 7th division coincides.
14. MSR = 0.2 cm, 8th division coincides.
15. MSR = 5.0 cm, 3rd division coincides.
16. MSR = 1.1 cm, 5th division coincides.
17. MSR = 4.6 cm, 9th division coincides.
18. MSR = 2.3 cm, 1st division coincides.
19. MSR = 6.7 cm, 4th division coincides.
20. MSR = 3.9 cm, 6th division coincides.
21. MSR = 0.6 cm, 2nd division coincides.
22. MSR = 9.0 cm, 7th division coincides.

Vernier — Zero Error & Correction

23. Zero error = +0.02 cm. MSR = 2.3 cm, 5th division coincides.
24. Zero error = -0.03 cm. MSR = 1.5 cm, 4th division coincides.
25. Zero error = +0.05 cm. MSR = 4.0 cm, 6th division coincides.

- 26. Zero error = -0.02 cm. MSR = 0.9 cm, 9th division coincides.
- 27. Zero error = $+0.01$ cm. MSR = 5.5 cm, 9th division coincides.
- 28. Zero error = -0.05 cm. MSR = 0.0 cm, 3rd division coincides.
- 29. Zero error = $+0.04$ cm. MSR = 2.2 cm, 2nd division coincides.
- 30. Zero error = -0.01 cm. MSR = 3.6 cm, 5th division coincides.
- 31. Zero error = $+0.03$ cm. MSR = 1.0 cm, 1st division coincides.
- 32. Zero error = -0.04 cm. MSR = 6.1 cm, 8th division coincides.
- 33. Zero error = $+0.06$ cm. MSR = 2.9 cm, 3rd division coincides.
- 34. Zero error = -0.06 cm. MSR = 0.4 cm, 7th division coincides.
- 35. Zero error = $+0.02$ cm. MSR = 7.0 cm, 4th division coincides.
- 36. Zero error = -0.03 cm. MSR = 1.8 cm, 2nd division coincides.
- 37. Jaws closed: 3rd division coincides, right of zero. MSR = 2.0 cm, 8th division coincides.
- 38. Jaws closed: 7th division (of 10) coincides, left of zero. MSR = 3.3 cm, 6th division coincides.
- 39. Jaws closed: 9th division (of 10) coincides, left of zero. MSR = 3.1 cm, 2nd division coincides.
- 40. Jaws closed: 6th division (of 10) coincides, left of zero. MSR = 2.2 cm, 1st division coincides.
- 41. Jaws closed: 5th division coincides, right of zero. MSR = 4.4 cm, 7th division coincides.
- 42. Jaws closed: 2nd division coincides, right of zero. MSR = 5.0 cm, 9th division coincides.
- 43. Jaws closed: 8th division (of 10) coincides, left of zero. MSR = 1.7 cm, 3rd division coincides.
- 44. Jaws closed: 4th division coincides, right of zero. MSR = 6.6 cm, 5th division coincides.
- 45. Jaws closed: 1st division (of 10) coincides, left of zero. MSR = 0.5 cm, 2nd division coincides.

Screw Gauge — Pitch & Least Count

- 46. Pitch = 1 mm, 100 divisions.
- 47. Pitch = 0.5 mm, 50 divisions.
- 48. Advances 5 mm in 5 rotations, 100 divisions.
- 49. Advances 3 mm in 6 rotations, 50 divisions.

- 50. Advances 4 mm in 8 rotations, 100 divisions.
- 51. Pitch = 1 mm, 50 divisions.
- 52. Advances 1 mm in 2 rotations, 50 divisions.
- 53. Pitch = 1 mm, 200 divisions.
- 54. Advances 2 mm in 2 rotations, 100 divisions.
- 55. Advances 6 mm in 12 rotations, 50 divisions.
- 56. Pitch = 2 mm, 100 divisions.
- 57. Advances 10 mm in 20 rotations, 100 divisions.

Screw Gauge — Readings

- 58. MSR = 3 mm, 28th division.
- 59. MSR = 5 mm, 45th division.
- 60. MSR = 0 mm, 12th division.
- 61. MSR = 7 mm, 67th division.
- 62. MSR = 2 mm, 5th division.
- 63. MSR = 4 mm, 90th division.
- 64. MSR = 1 mm, 99th division.
- 65. MSR = 6 mm, 33rd division.
- 66. MSR = 8 mm, 8th division.
- 67. MSR = 3 mm, 50th division.
- 68. MSR = 9 mm, 15th division.
- 69. MSR = 2 mm, 72nd division.
- 70. MSR = 5 mm, 4th division.
- 71. MSR = 0 mm, 85th division.
- 72. MSR = 6 mm, 21st division.
- 73. MSR = 10 mm, 60th division.
- 74. MSR = 4 mm, 3rd division.
- 75. MSR = 7 mm, 95th division.
- 76. MSR = 1 mm, 18th division.
- 77. MSR = 3 mm, 77th division.

Screw Gauge — Zero Error & Correction

- 78.** Zero error = +0.03 mm. MSR = 4 mm, 67th division.
- 79.** Zero error = +0.02 mm. MSR = 5 mm, 20th division.
- 80.** Zero error = +0.01 mm. MSR = 0 mm, 52nd division.
- 81.** Zero error = +0.04 mm. MSR = 6 mm, 8th division.
- 82.** Zero error = −0.03 mm. MSR = 2 mm, 44th division.
- 83.** Zero error = −0.01 mm. MSR = 3 mm, 12th division.
- 84.** Zero error = +0.05 mm. MSR = 1 mm, 30th division.
- 85.** Zero error = −0.02 mm. MSR = 7 mm, 9th division.
- 86.** Jaws closed: 95th division on reference line (above datum). MSR = 2 mm, 36th division.
- 87.** Jaws closed: 96th division on reference line (above datum). MSR = 3 mm, 10th division.
- 88.** Jaws closed: 98th division on reference line (above datum). MSR = 1 mm, 15th division.
- 89.** Jaws closed: 4th division on reference line (below datum). MSR = 5 mm, 25th division.
- 90.** Jaws closed: 97th division on reference line (above datum). MSR = 4 mm, 60th division.
- 91.** Jaws closed: 2nd division on reference line (below datum). MSR = 6 mm, 50th division.

Screw Gauge — Half-Millimetre Pitch (pitch = 0.5 mm, 50 divisions, LC = 0.01 mm)

- 92.** 7 mm mark exposed, 7.5 mm half-mark exposed, 12th division on reference line.
- 93.** 3 mm mark exposed, 3.5 mm half-mark not exposed, 45th division on reference line.
- 94.** 5 mm mark exposed, 5.5 mm half-mark exposed, 8th division on reference line.
- 95.** 10 mm mark exposed, 10.5 mm half-mark not exposed, 30th division on reference line.
- 96.** 2 mm mark exposed, 2.5 mm half-mark exposed, 5th division on reference line.
- 97.** 8 mm mark exposed, 8.5 mm half-mark not exposed, 22nd division on reference line.
- 98.** 4 mm mark exposed, 4.5 mm half-mark exposed, 40th division on reference line.

- 99.** 1 mm mark exposed, 1.5 mm half-mark not exposed, 10th division on reference line.
- 100.** 6 mm mark exposed, 6.5 mm half-mark exposed, 33rd division on reference line.

Answer Key

Vernier Calipers & Screw Gauge — Final Answers

Vernier — Readings

1. 1.24 cm
2. 2.57 cm
3. 0.89 cm
4. 3.42 cm
5. 5.68 cm
6. 0.53 cm
7. 6.26 cm
8. 4.11 cm
9. 2.05 cm
10. 7.39 cm
11. 1.96 cm
12. 3.02 cm
13. 8.47 cm
14. 0.28 cm
15. 5.03 cm
16. 1.15 cm
17. 4.69 cm
18. 2.31 cm
19. 6.74 cm
20. 3.96 cm
21. 0.62 cm
22. 9.07 cm

Vernier — Zero Error & Correction

23. 2.33 cm
24. 1.57 cm
25. 4.01 cm

- 26. 1.01 cm
- 27. 5.58 cm
- 28. 0.08 cm
- 29. 2.18 cm
- 30. 3.66 cm
- 31. 0.98 cm
- 32. 6.22 cm
- 33. 2.87 cm
- 34. 0.53 cm
- 35. 7.02 cm
- 36. 1.85 cm
- 37. ZE = +0.03 cm; Corrected = 2.05 cm
- 38. ZE = -0.03 cm; Corrected = 3.39 cm
- 39. ZE = -0.01 cm; Corrected = 3.13 cm
- 40. ZE = -0.04 cm; Corrected = 2.25 cm
- 41. ZE = +0.05 cm; Corrected = 4.42 cm
- 42. ZE = +0.02 cm; Corrected = 5.07 cm
- 43. ZE = -0.02 cm; Corrected = 1.75 cm
- 44. ZE = +0.04 cm; Corrected = 6.61 cm
- 45. ZE = -0.09 cm; Corrected = 0.61 cm

Screw Gauge — Pitch & Least Count

- 46. 0.01 mm
- 47. 0.01 mm
- 48. Pitch = 1 mm; LC = 0.01 mm
- 49. Pitch = 0.5 mm; LC = 0.01 mm
- 50. Pitch = 0.5 mm; LC = 0.005 mm
- 51. 0.02 mm
- 52. Pitch = 0.5 mm; LC = 0.01 mm
- 53. 0.005 mm
- 54. Pitch = 1 mm; LC = 0.01 mm
- 55. Pitch = 0.5 mm; LC = 0.01 mm

56. 0.02 mm
57. Pitch = 0.5 mm; LC = 0.005 mm

Screw Gauge — Readings

58. 3.28 mm
59. 5.45 mm
60. 0.12 mm
61. 7.67 mm
62. 2.05 mm
63. 4.90 mm
64. 1.99 mm
65. 6.33 mm
66. 8.08 mm
67. 3.50 mm
68. 9.15 mm
69. 2.72 mm
70. 5.04 mm
71. 0.85 mm
72. 6.21 mm
73. 10.60 mm
74. 4.03 mm
75. 7.95 mm
76. 1.18 mm
77. 3.77 mm

Screw Gauge — Zero Error & Correction

78. 4.64 mm
79. 5.18 mm
80. 0.51 mm
81. 6.04 mm
82. 2.47 mm
83. 3.13 mm

- 84. 1.25 mm
- 85. 7.11 mm
- 86. ZE = -0.05 mm; Corrected = 2.41 mm
- 87. ZE = -0.04 mm; Corrected = 3.14 mm
- 88. ZE = -0.02 mm; Corrected = 1.17 mm
- 89. ZE = $+0.04$ mm; Corrected = 5.21 mm
- 90. ZE = -0.03 mm; Corrected = 4.63 mm
- 91. ZE = $+0.02$ mm; Corrected = 6.48 mm

Screw Gauge — Half-Millimetre Pitch (pitch = 0.5 mm, 50 divisions, LC = 0.01 mm)

- 92. 7.62 mm
- 93. 3.45 mm
- 94. 5.58 mm
- 95. 10.30 mm
- 96. 2.55 mm
- 97. 8.22 mm
- 98. 4.90 mm
- 99. 1.10 mm
- 100. 6.83 mm